MUBAS

SCHOOL OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

MID SEMESTER EXAM

DATA STRUCTURES AND ALGORITHMS (CIS-DSA-211)

Iterative Algorithms, sorting algorithms, stacks and queue simulations

(80 Marks)

Lecture: A. Taylor

7th September 2026

Instructions:

Answer all questions.

1. What is an iterative algorithm? (5 marks)

2. Count and sum large sales (20 marks)
   A shop records daily sales amounts in the array below:
   Sales = [120, 75, 200, 125, 40, 55, 60, 150, 80, 90, 300]
   Write an Iterative algorithm that calculates the frequency of sales – choose appropriate bins.

   a. Write the algorithm in pseudocode. (7 marks)
   b. Perform a dry run using a trace table containing I, sales[i], condition, count and total. (5 marks)
   c. Give the final values of count and total. (3 marks)
   d. Determine the time complexity. (2 marks)
   e. Explain whether the complexity changes if the array contains 10,000 elements. (3 marks)

3. Write the pseudocode for bubble sort. (10 marks)

4. Use selection sort to sort an array into ascending order. (20 marks)
   A = [7, 3, 6, 3, 2, 6, 5]
   a. Write the complete selection sort pseudocode. (7 marks)

b. Construct a trace table showing pass, current position, index of minimum, minimum value and array after swap. (7 marks)
c. Show every pass and state the final sorted array. (3 marks
d. State and justify the worst-case time complexity. (3marks)

5. Bank service queue (25 marks)
   A bank used two queues: customer queue for customers waiting to see a teller, and transaction queue for accepted transactions waiting to be processed. A customer arrives every 3 minutes. It takes the teller 1 minute to receive a transaction. The processing system handles one transaction at a time.

Transaction	Processing time
Withdrawal	3 minutes
Deposit	4 minutes
Transfer	6 minutes

Customer	Arrival	Transaction
Aisha	0	Transfer
Blessings	3	Withdrawal
Charles	6	Deposit
Diana	9	Withdrawal
Edward	12	Transfer

a. Write the pseudocode to simulate the bank from minute 0 to minute 20. (8 marks)
b. Construct a trace table showing time. Arriving customer, customerQueue, customer at teller, transactionQueue, current transaction and finish time. (8 marks)
c. State when Aisha's and Blessing's transactions are completed. (3 marks)
d. Identify when transactionQueue starts accumulating transactions. (3 marks)
e. State which transactions have not been completed at minute 20 and explain why FIFO is appropriate. (3 marks)